

Response to Reviewers and Editor

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Title: OphDDI-CDS: An Absorption-First Deterministic Cascade for Ophthalmic Drug–Drug Interaction Screening

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Decision: Revise

Dear Dr Shelke and Dr Parikesit,

We thank the editor for a careful assessment that identified three concrete, actionable gaps. All three have been addressed, and the manuscript has also been revised in line with the general guidance to report results accurately, remove overstated conclusions, and explain limitations fully. Below we respond point by point; the changes are marked in the revised manuscript.

We also record one change to the framing of the paper, made in response to the editorial guidance about overstated conclusions. It is described under “Changes to framing and claims” below, because it affects the abstract and conclusion and we want the reviewers to be able to assess it deliberately.

Editor comment 1 — Source code must be deposited in an open repository

“After I did thorough examination of the manuscript, including on the supplementary material, I can’t find any pointers to the source code repository. So it is not feasible on my end to repeat the computational procedure. Therefore, the source code of the computational procedure should be deposited in the Github or similar open source repository, so repeatability is possible.”

Accepted in full. The editor is right, and the omission was ours: the previous submission described a “downloadable release package” as supplementary material without ever pointing to a public, versioned repository. The complete source code is now deposited at:

<https://github.com/wanguoqing932/ophddi-cds> (public; accessible without a GitHub account)

The repository contains the production screening engine, the curated knowledge base, the rule-citation verification tooling, the evaluation scripts, and a test suite. As requested, the deposited code is complete and runnable, not a stub:

Component	Contents
src/ophthalmic_ddi_cds_agent/	Screening engine, registry layer, rule matcher, citation sources
configs/	rules.yaml (40 rules + 3 patient-factor

Component	Contents
	policies), class_matrix.yaml, drug_classes.yaml, flag_catalog.yaml
data/curation/	Candidate registries and the rule-to-citation map
data/raw/evidence/citations/	Cached Crossref / OpenAlex / PubMed responses, so citation verification replays offline
scripts/	Demo, evaluation, and build scripts
tests/	Test suite — 34 passed, 1 skipped

We verified the deposit the way a reviewer would: cloning the public repository into a clean directory and running it there, rather than testing only in our own working copy. The clone produced 96 files and the demonstration completed with all 25 mock cases reproducing their expected risk level.

Every number in the manuscript is reproducible from this deposit. The deterministic cascade described in the paper — which is the contribution — requires no API key and no network access, so the editor can reproduce the system’s own results directly. The LLM-baseline comparisons are also included but require the reviewer to supply their own model API credentials; this is stated in the repository README rather than left implicit.

Editor comment 2 — Deposit mock-up data so the pipeline can be inspected

“Pertaining the data, you don’t need to deposited your computed data in the repository, and you can proceed by depositing mock up data. So the reviewer can really check whether your pipeline is working or not. Real data could be deposited later, if this manuscript is accepted.”

Accepted. The repository ships a self-contained mock dataset and a single command that exercises the full pipeline:

```
python scripts/demo_pipeline.py
```

This runs 25 drug pairs through the production cascade with no API key and no network access, and checks each returned risk level against the value frozen at release. The mock set deliberately covers all three decision paths the paper describes — rule match (13 cases), class-matrix match (11 cases), and the unknown-drug fail-safe path (1 case) — so a reviewer can confirm not only that the code runs but that the three documented layers each behave as claimed. The command also verifies the fail-safe property asserted in Section 2.2: a medication outside the registry returns an explicit unknown-drug path and receives **no** risk grade, rather than a default.

25/25 mock cases reproduced their expected level
PASS: no risk grade is assigned to an unknown drug
All checks passed.

data/mock/mock_evidence_chunks.jsonl provides eight illustrative evidence passages for the retrieval path. These are explicitly labelled mock-synthetic in the file and in the README —

they are written for the repository and are **not real citations** — so that no reader mistakes demonstration text for evidence.

We note for transparency that the two entity registries (data/seed/entities_a.csv, entities_b.csv) are deposited as **real** data rather than mock, because they contain only drug names, drug classes, mechanism flags and absorption tiers — no patient data and no personal information. Depositing the real registries means the demo runs against the same 113×232 agent set used in the manuscript, which we judged more useful to a reviewer than a synthetic registry. We will of course replace them with mock registries if the editor prefers.

Editor comment 3 — Ethics waiver opinion must appear in the ethics statement

“As this manuscript involves expert opinion from health care professionals, waiver opinion for the ethics approval should be provided in the ethics statement.”

Accepted, and the statement has been rewritten. The previous version stated merely that “ethics approval was not required for this study design under applicable institutional policy” and offered a waiver “on request” — which is exactly the weakness the editor identified. The revised statement now sets out the basis for the waiver explicitly:

This study did not involve human participants, human tissue, patient data, or clinical interventions. The only human involvement was an expert-opinion exercise in which two ophthalmologists independently rated 12 hypothetical drug-pair vignettes constructed by the study team; no identifiable information was collected, and both experts provided informed consent for their anonymised ratings to be used in research and publication. Under the policy of the Ethics Committee of the First Affiliated Hospital of Xinjiang Medical University, a study of this type (expert opinion on hypothetical vignettes, with no human participants and no identifiable data) does not require full ethics review; a formal waiver opinion has been requested from the committee and is provided as supplementary material. The work was conducted in accordance with the Declaration of Helsinki.

We have also added the two other missing elements the journal requires in this section: a Consent for publication subheading (“Not applicable — this study did not involve data from any individual person”) and an Acknowledgements subheading.

Note to the editor, should it be useful: the study described here is distinct from the expert-opinion exercise, so no approval number for a separate protocol is cited in the ethics statement. The waiver is documented on the basis of the study design described above.

Changes to framing and claims (editorial guidance)

The editor asked that results be reported accurately, overstated conclusions rewritten, and limitations fully explained. Reviewing the manuscript against its own evidence, we found that the weakest-supported claim was the one leading the paper, and we have restructured accordingly. No data, analysis, or number has changed; the framing has.

What changed.

1. **The abstract no longer leads with the 0.915 accuracy figure.** That figure measures agreement with a gold standard that the authors themselves constructed from the same literature sources that informed the rules — a homogeneous construction we already disclosed. It is now reported as what it is, the internal consistency of an explicitly evidence-graded encoding, and the paper's opening claim is instead the measurement that does not depend on the gold at all: across all 1,933 registry-matched DDInter pairs, absorption-aware assessment grades 75.0% of pairs lower, with 77.0% of that shift attributable to a single mechanism, low systemic absorption from the topical route.
2. **That registry-scale audit has been promoted to its own Results subsection and moved ahead of the gold-dependent audit.** It is now presented as the primary quantitative finding, with its two boundary conditions stated in the Results rather than deferred: it quantifies divergence and not error, and the absorption tiers that drive it were assigned by the study team from pharmacokinetic sources and regulatory labels.
3. **The expert assessment is now framed as a finding rather than a failure of validation.** Two ophthalmologists agreed closely with each other ($\kappa = 0.846$) but graded most cases one tier below both the cascade and its gold. Crucially, their agreement with the gold was numerically identical to their agreement with the system ($\kappa = 0.125$ and 0.010), because the cascade reproduced its gold on all 12 cases. The divergence therefore lies between clinical prior and evidence-graded assessment, not in the system. We state plainly that with $n = 12$ it cannot be distinguished from chance-level agreement.
4. **The Discussion now states the methodological implication directly:** a severity rating exported from a systemic-route database is not a valid ophthalmic ground truth, and evaluating an ophthalmic tool against one measures the distance between two assumptions rather than accuracy. This is why the unadjusted-label agreement figures (0.210 and 0.443, Section 3.2) are reported as a quantification of the adjustment rather than as performance.
5. **The conclusion was rewritten** to lead with what the study establishes about the grading of ophthalmic DDI risk, and to state the audit-set high-risk sensitivity (0.440) as the number that bounds the system's current value as a safety screen.

Limitations. Section 3.8 and Section 4.4 have been revised to state the following without softening: the L1 gold is literature-derived and author-annotated rather than expert-adjudicated, and shares its sources with the rules; only the 40 procedurally frozen cases are independent in a procedural sense, and only 20 of those were free of development exposure; 78 of 118 L1 cases overlap the development-phase validation set by case identity; the audit gold inherits the system's absorption assumption for 30 of 92 cases; audit-set high-risk sensitivity is 0.440, the lowest of the four methods; and at 1% prevalence the audit operating point implies a positive predictive value of approximately 9%, so the system's value lies in systematic coverage and traceability rather than in screening yield.

We are grateful for the editor's engagement with the supplement, which is what surfaced the missing repository pointer.

Yours sincerely,

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